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Continuous glucose monitoring system

Abstract

Methods and systems for delaying alarms that include detecting an analyte level using an analyte sensor; and delaying the annunciation of an analyte alarm after the analyte level crosses an analyte threshold, wherein the delay is based on one or both of (1) a magnitude of difference between the analyte level and the analyte threshold and (2) a duration of time in which the analyte level has crossed the analyte threshold.

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7179226	12/2006	Crothall et al.	N/A	N/A
7183102	12/2006	Monfre et al.	N/A	N/A

7190988	12/2006	Say et al.	N/A	N/A
7192450	12/2006	Brauker et al.	N/A	N/A
7198606	12/2006	Boecker et al.	N/A	N/A
7225535	12/2006	Feldman et al.	N/A	N/A
7226442	12/2006	Sheppard et al.	N/A	N/A
7226978	12/2006	Tapsak et al.	N/A	N/A
7241266	12/2006	Zhou et al.	N/A	N/A
7258673	12/2006	Racchini et al.	N/A	N/A
7267665	12/2006	Steil et al.	N/A	N/A
7276029	12/2006	Goode, Jr. et al.	N/A	N/A
7278983	12/2006	Ireland et al.	N/A	N/A
7286894	12/2006	Grant et al.	N/A	N/A
7295867	12/2006	Berner et al.	N/A	N/A
7299082	12/2006	Feldman et al.	N/A	N/A
7301463	12/2006	Paterno	N/A	N/A
7310544	12/2006	Brister et al.	N/A	N/A
7317938	12/2007	Lorenz et al.	N/A	N/A
7335294	12/2007	Heller et al.	N/A	N/A
7344500	12/2007	Talbot et al.	N/A	N/A
7354420	12/2007	Steil et al.	N/A	N/A
7364592	12/2007	Carr-Brendel et al.	N/A	N/A
7366556	12/2007	Brister et al.	N/A	N/A
7379765	12/2007	Petisce et al.	N/A	N/A
7399277	12/2007	Saidara et al.	N/A	N/A
7401111	12/2007	Batman et al.	N/A	N/A
7402153	12/2007	Steil et al.	N/A	N/A
7424318	12/2007	Brister et al.	N/A	N/A
7460898	12/2007	Brister et al.	N/A	N/A
7467003	12/2007	Brister et al.	N/A	N/A
7468125	12/2007	Kraft et al.	N/A	N/A
7471972	12/2007	Rhodes et al.	N/A	N/A
7474992	12/2008	Ariyur	N/A	N/A
7492254	12/2008	Bandy et al.	N/A	N/A
7494465	12/2008	Brister et al.	N/A	N/A
7497827	12/2008	Brister et al.	N/A	N/A
7499002	12/2008	Blasko et al.	N/A	N/A
7519408	12/2008	Rasdal et al.	N/A	N/A
7519478	12/2008	Bartkowiak et al.	N/A	N/A
7523004	12/2008	Bartkowiak et al.	N/A	N/A
7547281	12/2008	Hayes et al.	N/A	N/A
7569030	12/2008	Lebel et al.	N/A	N/A
7577469	12/2008	Aronowitz et al.	N/A	N/A
7583990	12/2008	Goode, Jr. et al.	N/A	N/A
7591801	12/2008	Brauker et al.	N/A	N/A
7599726	12/2008	Goode, Jr. et al.	N/A	N/A
7613491	12/2008	Boock et al.	N/A	N/A
7615007	12/2008	Shults et al.	N/A	N/A
7618369	12/2008	Hayter et al.	N/A	N/A
7620438	12/2008	He	N/A	N/A
7630748	12/2008	Budiman	N/A	N/A

7632228	12/2008	Brauker et al.	N/A	N/A
7635594	12/2008	Holmes et al.	N/A	N/A
7637868	12/2008	Saint et al.	N/A	N/A
7640048	12/2008	Dobbles et al.	N/A	N/A
7651596	12/2009	Petisce et al.	N/A	N/A
7651845	12/2009	Doyle, III et al.	N/A	N/A
7654956	12/2009	Brister et al.	N/A	N/A
7657297	12/2009	Simpson et al.	N/A	N/A
7670263	12/2009	Ellis et al.	N/A	N/A
7684872	12/2009	Carney et al.	N/A	N/A
7699775	12/2009	Desai et al.	N/A	N/A
7699964	12/2009	Feldman et al.	N/A	N/A
7711402	12/2009	Shults et al.	N/A	N/A
7711493	12/2009	Bartkowiak et al.	N/A	N/A
7713574	12/2009	Brister et al.	N/A	N/A
7715893	12/2009	Kamath et al.	N/A	N/A
7736310	12/2009	Taub et al.	N/A	N/A
7751864	12/2009	Buck, Jr.	N/A	N/A
7766829	12/2009	Sloan et al.	N/A	N/A
7768387	12/2009	Fennell et al.	N/A	N/A
7774145	12/2009	Bruaker et al.	N/A	N/A
7778680	12/2009	Goode, Jr. et al.	N/A	N/A
7785256	12/2009	Koh	N/A	N/A
7811231	12/2009	Jin et al.	N/A	N/A
7813809	12/2009	Strother et al.	N/A	N/A
7826382	12/2009	Sicurello et al.	N/A	N/A
7826981	12/2009	Goode et al.	N/A	N/A
7842174	12/2009	Zhou et al.	N/A	N/A
7857760	12/2009	Brister et al.	N/A	N/A
7885697	12/2010	Brister et al.	N/A	N/A
7885698	12/2010	Feldman et al.	N/A	N/A
7889069	12/2010	Fifolt et al.	N/A	N/A
7894869	12/2010	Hoarau	N/A	N/A
7899511	12/2010	Shults et al.	N/A	N/A
7899545	12/2010	John	N/A	N/A
7905833	12/2010	Brister et al.	N/A	N/A
7912655	12/2010	Power et al.	N/A	N/A
7912674	12/2010	Killoren Clark et al.	N/A	N/A
7914450	12/2010	Goode, Jr. et al.	N/A	N/A
7920907	12/2010	McGarraugh et al.	N/A	N/A
7928850	12/2010	Hayter et al.	N/A	N/A
7938797	12/2010	Estes	N/A	N/A
7941200	12/2010	Weinert et al.	N/A	N/A
7946984	12/2010	Brister et al.	N/A	N/A
7946985	12/2010	Mastrototaro et al.	N/A	N/A
7972296	12/2010	Braig et al.	N/A	N/A
7974672	12/2010	Shults et al.	N/A	N/A
7976466	12/2010	Ward et al.	N/A	N/A
7976467	12/2010	Young et al.	N/A	N/A

7978063	12/2010	Baldus et al.	N/A	N/A
7996158	12/2010	Hayter et al.	N/A	N/A
8005524	12/2010	Brauker et al.	N/A	N/A
8010174	12/2010	Goode et al.	N/A	N/A
8010256	12/2010	Oowada	N/A	N/A
8060173	12/2010	Goode, Jr. et al.	N/A	N/A
8066639	12/2010	Nelson et al.	N/A	N/A
8073008	12/2010	Mehta	370/468	A61M 5/142
8103471	12/2011	Hayter	N/A	N/A
8140312	12/2011	Hayter et al.	N/A	N/A
8160669	12/2011	Brauker et al.	N/A	N/A
8160900	12/2011	Taub et al.	N/A	N/A
8170803	12/2011	Kamath et al.	N/A	N/A
8190223	12/2011	Al-Ali et al.	N/A	N/A
8192394	12/2011	Estes et al.	N/A	N/A
8207859	12/2011	Enegren et al.	N/A	N/A
8216138	12/2011	McGarraugh et al.	N/A	N/A
8216139	12/2011	Brauker et al.	N/A	N/A
8239166	12/2011	Hayter et al.	N/A	N/A
8255026	12/2011	Al-Ali	N/A	N/A
8260558	12/2011	Hayter et al.	N/A	N/A
8282549	12/2011	Brauker et al.	N/A	N/A
8374667	12/2012	Brauker et al.	N/A	N/A
8374668	12/2012	Hayter et al.	N/A	N/A
8376945	12/2012	Hayter et al.	N/A	N/A
8377271	12/2012	Mao et al.	N/A	N/A
8409093	12/2012	Bugler	N/A	N/A
8444560	12/2012	Hayter et al.	N/A	N/A
8461985	12/2012	Fennell et al.	N/A	N/A
8478406	12/2012	Burnes et al.	N/A	N/A
8478557	12/2012	Hayter et al.	N/A	N/A
8484005	12/2012	Hayter et al.	N/A	N/A
8497777	12/2012	Harper	N/A	N/A
8514086	12/2012	Harper et al.	N/A	N/A
8543354	12/2012	Luo et al.	N/A	N/A
8560038	12/2012	Hayter et al.	N/A	N/A
8567393	12/2012	Hickle et al.	N/A	N/A
8571808	12/2012	Hayter	N/A	N/A
8583205	12/2012	Budiman et al.	N/A	N/A
8597274	12/2012	Sloan et al.	N/A	N/A
8597570	12/2012	Terashima et al.	N/A	N/A
8600681	12/2012	Hayter et al.	N/A	N/A
8608923	12/2012	Zhou et al.	N/A	N/A
8612163	12/2012	Hayter et al.	N/A	N/A
8657746	12/2013	Roy	N/A	N/A
8710993	12/2013	Hayter et al.	N/A	N/A
8730058	12/2013	Harper	N/A	N/A
8734422	12/2013	Hayter	N/A	N/A
8816862	12/2013	Harper et al.	N/A	N/A
8821792	12/2013	Drucker et al.	N/A	N/A

8834366	12/2013	Hayter et al.	N/A	N/A
8845536	12/2013	Brauker et al.	N/A	N/A
9000914	12/2014	Baker et al.	N/A	N/A
9060719	12/2014	Hayter et al.	N/A	N/A
9119528	12/2014	Cobelli	N/A	N/A
9143569	12/2014	Mensinger et al.	N/A	N/A
9178752	12/2014	Harper	N/A	N/A
9186113	12/2014	Harper et al.	N/A	N/A
9215980	12/2014	Tran et al.	N/A	N/A
9226714	12/2015	Harper et al.	N/A	N/A
9289179	12/2015	Hayter et al.	N/A	N/A
9339197	12/2015	Griswold et al.	N/A	N/A
9398872	12/2015	Hayter et al.	N/A	N/A
9408566	12/2015	Hayter et al.	N/A	N/A
9414775	12/2015	Colvin, Jr. et al.	N/A	N/A
9439586	12/2015	Bugler	N/A	N/A
9446194	12/2015	Kamath et al.	N/A	N/A
9452258	12/2015	Dobbles et al.	N/A	N/A
9483608	12/2015	Hayter et al.	N/A	N/A
9549694	12/2016	Harper et al.	N/A	N/A
9558325	12/2016	Hayter et al.	N/A	N/A
9717421	12/2016	Griswold et al.	N/A	N/A
9743872	12/2016	Hayter et al.	N/A	N/A
9801541	12/2016	Mensinger et al.	N/A	N/A
9804148	12/2016	Hayter et al.	N/A	N/A
9814400	12/2016	Cendrillon et al.	N/A	N/A
9814416	12/2016	Harper et al.	N/A	N/A
9833181	12/2016	Hayter et al.	N/A	N/A
9980140	12/2017	Spencer et al.	N/A	N/A
10009244	12/2017	Harper et al.	N/A	N/A
10123752	12/2017	Harper et al.	N/A	N/A
10159432	12/2017	Pathak et al.	N/A	N/A
10209817	12/2018	Cazzoli et al.	N/A	N/A
10264971	12/2018	Kennedy et al.	N/A	N/A
10375222	12/2018	Mandapaka et al.	N/A	N/A
10456091	12/2018	Harper et al.	N/A	N/A
10702215	12/2019	Hampapuram et al.	N/A	N/A
10772572	12/2019	Harper et al.	N/A	N/A
10855788	12/2019	Arabo et al.	N/A	N/A
11213204	12/2021	Mensinger et al.	N/A	N/A
11241175	12/2021	Sloan et al.	N/A	N/A
11991175	12/2023	Rolfe et al.	N/A	N/A
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2009/0105571	12/2008	Fennell et al.	N/A	N/A
2009/0105636	12/2008	Hayter et al.	N/A	N/A
2009/0112154	12/2008	Montgomery et al.	N/A	N/A
2009/0112478	12/2008	Mueller, Jr. et al.	N/A	N/A
2009/0112626	12/2008	Talbot et al.	N/A	N/A
2009/0114811	12/2008	Landgraf	N/A	N/A
2009/0124877	12/2008	Goode et al.	N/A	N/A
2009/0124878	12/2008	Goode et al.	N/A	N/A
2009/0124879	12/2008	Brister et al.	N/A	N/A
2009/0124964	12/2008	Leach et al.	N/A	N/A
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2009/0137886	12/2008	Shariati et al.	N/A	N/A
2009/0137887	12/2008	Shariati et al.	N/A	N/A
2009/0143659	12/2008	Li et al.	N/A	N/A
2009/0143660	12/2008	Brister et al.	N/A	N/A
2009/0143725	12/2008	Peyser et al.	N/A	N/A
2009/0149717	12/2008	Brauer et al.	N/A	N/A
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2009/0156919	12/2008	Brister et al.	N/A	N/A
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2009/0163790	12/2008	Brister et al.	N/A	N/A
2009/0163791	12/2008	Brister et al.	N/A	N/A
2009/0163855	12/2008	Shin et al.	N/A	N/A
2009/0164190	12/2008	Hayter	N/A	N/A
2009/0164239	12/2008	Hayter et al.	N/A	N/A
2009/0164251	12/2008	Hayter	N/A	N/A
2009/0177068	12/2008	Stivoric et al.	N/A	N/A
2009/0178459	12/2008	Li et al.	N/A	N/A

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2009/0192366	12/2008	Mensinger et al.	N/A	N/A
2009/0192380	12/2008	Shariati et al.	N/A	N/A
2009/0192722	12/2008	Shariati et al.	N/A	N/A
2009/0192724	12/2008	Brauker et al.	N/A	N/A
2009/0192745	12/2008	Kamath et al.	N/A	N/A
2009/0192751	12/2008	Kamath et al.	N/A	N/A
2009/0198118	12/2008	Hayter et al.	N/A	N/A
2009/0203981	12/2008	Brauker et al.	N/A	N/A
2009/0204341	12/2008	Brauker et al.	N/A	N/A
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2009/0257911	12/2008	Thomas et al.	N/A	N/A
2009/0275815	12/2008	Bickoff et al.	N/A	N/A
2009/0275887	12/2008	Estes	N/A	N/A
2009/0287073	12/2008	Boock et al.	N/A	N/A
2009/0287074	12/2008	Shults et al.	N/A	N/A
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2010/0049164	12/2009	Estes	N/A	N/A
2010/0057040	12/2009	Hayter	N/A	N/A
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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS (1) The present application is a continuation of U.S. patent application Ser. No. 17/567,684, filed Jan. 3, 2022, which is a continuation of U.S. patent application Ser. No. 16/560,497, filed Sep. 4, 2019, now U.S. Pat. No. 11,213,622 which is a continuation of U.S. patent application Ser. No. 15/282,688 filed on Sep. 30, 2016, which is a divisional of U.S. patent application Ser. No. 14/076,109, filed Nov. 8, 2013, which is a continuation of U.S. patent application Ser. No. 12/785,144, filed May 21, 2010, now U.S. Pat. No. 8,597,274, which claims the benefit of U.S. Provisional Appl. No. 61/180,649, filed May 22, 2009, all of which are incorporated herein by reference in their entireties for all purposes. (2) This application is also related to U.S. application Ser. No. 12/784,981, filed May 21, 2010, entitled “Methods For Reducing False Hypoglycemia Alarm Occurrence,” and now U.S. Pat. No. 9,579,456 (U.S. Provisional No. 61/180,700, filed May 22, 2009); U.S. application Ser. No. 12/785,104, filed May 21, 2010, now U.S. Pat. No. 8,257,300, issued Sep. 4, 2012, entitled “Safety Features For Integrated Insulin Delivery System,” (U.S. Provisional Application No. 61/180,627, filed May 22, 2009); U.S. application Ser. No. 12/785,096, filed May 21, 2010, now U.S. Pat. No. 8,398,616, issued May 19, 2013, entitled “Safety Layer For Integrated Insulin Delivery System,” (U.S. Provisional Application No. 61/180,774, filed May 22, 2009); and U.S. application Ser. No. 12/785,196, filed May 21, 2010, entitled “Adaptive Insulin Delivery System,” now abandoned (U.S. Provisional Application No. 61/180,767, filed May 22, 2009).

BACKGROUND
(1) Diabetes is a metabolic disorder that afflicts tens of millions of people throughout the world. Diabetes results from the inability of the body to properly utilize and metabolize carbohydrates, particularly glucose. Normally, the finely tuned balance between glucose in the blood and glucose in bodily tissue cells is maintained by insulin, a hormone produced by the pancreas which controls, among other things, the transfer of glucose from blood into body tissue cells. Upsetting this balance causes many complications and pathologies including heart disease, coronary and peripheral artery sclerosis, peripheral neuropathies, retinal damage, cataracts, hypertension, coma, and death from

hypoglycemic shock.

(2) In persons with insulin-dependent diabetes, the symptoms of the disease can be controlled by administering additional insulin (or other agents that have similar effects) by injection or by external or implantable insulin pumps. The “correct” insulin dosage is a function of the level of glucose in the blood. Ideally, insulin administration should be continuously readjusted in response to changes in glucose level.

(3) Presently, systems are available for continuously monitoring a person's glucose levels by implanting a glucose sensitive probe into the person. Such probes measure various properties of blood or other tissues, including optical absorption, electrochemical potential and enzymatic products. The output of such sensors can be communicated to a hand held device or controller that is used to calculate an appropriate dosage of insulin to be delivered to the user of the continuous glucose monitor (CGM) in view of several factors, such as the user's present glucose level, insulin usage rate, carbohydrates consumed or to be consumed and exercise, among others. These calculations can then be used to control a pump that delivers the insulin, either at a controlled “basal” rate, or as a “bolus” into the user. When provided as an integrated system, the continuous glucose monitor, controller and pump work together to provide continuous glucose monitoring and insulin pump control.

(4) Such systems can be closed loop systems, where the amount of insulin being delivered is completely controlled by the controller and pump in conjunction with glucose level data received from the CGM device. Alternatively, such systems may be open loop systems, where the user evaluates the glucose level information from a glucose monitoring device and then instructs the pump accordingly, or the system may be a semi-closed loop system that combines various aspects of a closed loop and open loop system.

(5) Typically, present systems may be considered to be open or semi-closed loop in that they require intervention by a user to calculate and control the amount of insulin to be delivered. However, there may be periods when the user is not able to adjust insulin delivery. For example, when the user is sleeping, he or she cannot intervene in the delivery of insulin, yet control of a patient's glucose level is still necessary. A system capable of integrating and automating the functions of glucose monitoring and controlled insulin delivery into a closed loop system would be useful in assisting users in maintaining their glucose levels, especially during periods of the day when they are unable or unwilling to the required calculations to adjust insulin deliver to control their glucose level.

(6) What has been needed, and heretofore unavailable, is an integrated, automated system combining continuous glucose monitoring and controlled insulin delivery. Such a system would include various features to insure the accuracy of the glucose monitor and to protect the user from either under- or over-dosage of insulin. The system would include various functions for improving the usability, control, and safety of the system, including a variety of alarms which could be set by a user or a technician to avoid false alarms while ensuring adequate sensitivity to protect the user. The present invention satisfies these and other needs.

SUMMARY OF THE INVENTION

(7) Briefly, and in general terms, the invention is directed to new and improved systems and methods for management of a user's glucose level, including systems and methods for improving the usability and safety of such systems including continuous glucose monitors and a drug delivery pumps.

(8) In one aspect, the invention is directed to a system and method for use when insulin delivery is to be restarted after an unexpected stop in delivery. In this aspect of the invention, insulin history and glucose level history are used for calculating and recommending a bolus volume of insulin to be delivered to bring a user's insulin on board, that is, residual insulin that is unmetabolised and circulating in the user's blood stream since insulin was last added, up to the level it would have been had insulin delivery not been stopped.

(9) In another aspect, a safe amount and duration of basal rate insulin delivery is determined using model-based calculation of insulin on board, past insulin delivery information, past and present glucose information of the user and a predicted glucose profile over a near-future time horizon, among other factors, in the event that a closed loop insulin delivery system terminates closed loop operation unexpectedly.
(10) In still another aspect, the invention includes a system and method for enhancing alarm avoidance with pre-emptive pump control command modification, which is advantageous when compared with simply muting an alarm, and is less annoying than a persistent projected low glucose alarm. In yet another aspect, a user interface is provided that integrates and cooperates with the enhanced alarm avoidance system and method to facilitate use of the system.
(11) In a further aspect, a system and method is included that is directed to reducing the frequency of false low glucose alarms, particularly when an insulin delivery system is operating in a closed loop mode, by allowing for the possibility of the complete or partial recovery from an artifact, such as a sensor drop out, to occur prior to an alarm presentation to a user of the system.
(12) In another aspect, a model-based calculation of the present and near-future values of best estimate and upper/lower bounds of glucose is made to adjust a tiered alarm mechanism in order to account for predicted hypoglycemic and hyperglycemic events.
(13) In yet another aspect, the invention includes a method for determining a bolus volume to be administered to make up for a cessation of basal delivery of insulin, comprising: determining an amount of insulin remaining in a user's body such that insulin remaining=amount of insulin delivered before cessation×(user's insulin action time minus the time since insulin was last delivered)/(user's insulin action time); and calculating a bolus delivery to equal the amount of basal delivery lost. In another aspect, determining an amount of insulin remaining includes using a model to estimate the amount of insulin remaining.
(14) In still another aspect, the invention includes a method of determining an insulin delivery rate when a closed loop insulin delivery system terminates unexpectedly, comprising: a) providing a glucose level and predicting a future glucose level in order to determine the appropriate insulin bolus for the latest control action, and assuming that this commanded is immediately followed by open loop control where the pre-programmed basal delivery amount will be in effect thereafter; b) analyzing the future glucose level using this latest bolus value plus a future basal rate to determine if the future glucose level is acceptable, and if so, waiting for a selected period of time and repeating a); c) if the glucose level of b) is not acceptable, computing an alternate maximum temporary basal delivery amount that is lower than the previously assumed basal rate, such that the predicted future glucose level is acceptable, and if so, providing a temporary basal command at the computed basal rate and duration, and predicting a future glucose level; and if so, waiting for a selected period of time and repeating a); d) if the predicted future glucose level is unacceptable, projecting a future glucose level using the lowest temporary basal delivery rate computable in step c) and a maximum duration that is shorter than the maximum delivery duration, and, if the future glucose level is acceptable, providing a temporary basal command including the rate and duration of along with a nominal bolus command and then wait for a selected period of time and repeat a); and f) reducing the bolus command by minimum bolus resolution and repeat step a) such that the combination of the reduced bolus and a suitable temporary basal rate results in an acceptable predicted glucose profile if calculating a lower alternate temporary basal rate or a lower and shorter temporary basal rate does not result in an acceptable future glucose level. In still another aspect, the method further comprises alerting the user to take action to counter the effect of excessive insulin if an acceptable glucose profile cannot be obtained in step f) above.
(15) In another aspect, the present invention includes a method for predicting insulin needs at a future time to avoid unwanted out of range alarms and adjusting current insulin delivery; comprising: providing a current glucose level; providing a future time when a predicted glucose level is required; determining the predicted level at the future time based upon the current glucose

level, a value for insulin on board, current bolus parameters and current parameters defining an acceptable range of glucose levels; and adjusting insulin delivery to maintain the glucose level in a target range.
(16) In yet another aspect, the present invention includes a method for reducing false hypoglycemic alarms in a system under closed loop control; comprising: a) providing a current glucose level; b) determining if the first level is below a threshold level and if so, providing a further glucose level at the expiration of a selected period of time; c) determining if the glucose level taken after the expiration of the selected period of time is below the threshold level, and if so, alerting the user of a low glucose condition if the time between providing the two glucose levels is greater than a selected duration of time, and if not, repeating c) until either the time since the current glucose level and the last further glucose level exceeds the selected duration of time or the last further glucose level is above the threshold level.
(17) In yet another aspect, the present invention includes a method of adjusting glucose level alarm thresholds and alarm enunciation delay times in a system using CGM and insulin delivery system information, comprising using a model based state estimation, determining a predicted future glucose level and alerting a user only if the predicted future glucose level falls outside of a predetermined acceptable range. In an alternative aspect, the model based state estimation is a Kalman filter. In still another alternative aspect, the model based state estimation assesses the likelihood that a CGM measurement that exceeds a high or low threshold is due to a true event. In yet another alternative aspect, the model based state estimation assesses the likelihood that a CGM measurement that exceeds a high or low threshold is due to a sensor artifact.
(18) In still another aspect, the likelihood is determined by comparing the difference between the latest CGM measurement and interstitial glucose computed by the model prior to the latest CGM measurement; and in a further aspect, the method comprises adjusting the glucose alarm enunciation by adjusting the threshold level and duration in which the CGM measurement exceeds a given threshold before an alarm is enunciated.
(19) In yet another aspect, where a decreasing CGM signal close to a low threshold alarm value that is not accompanied by a corresponding amount of insulin history will result in a low threshold alarm being made less sensitive by lowering the threshold value and/or increasing the duration of delay before the CGM signal results in enunciation of a low threshold alarm. In an alternative aspect, where a decreasing CGM signal close to a low threshold alarm value that is accompanied by an insulin history that should have generated a much lower CGM value than is measured results in a low threshold alarm being made more sensitive by increasing the threshold value and/or decreasing the duration of delay before the CGM signal results in the enunciation of a low threshold alarm. In still another alternative aspect, where an increasing CGM signal close to a high threshold alarm value that is not accompanied by an appreciable amount of insulin history results in a high threshold alarm being made more sensitive by lowering the threshold value and/or decreasing the duration of delay before the CGM signal results in the enunciation of a high threshold alarm. In yet another alternative aspect, where an increasing CGM signal close to a high threshold alarm value that is accompanied by an appreciable amount of insulin history results in a high threshold alarm being made less sensitive by increasing the threshold value and/or increasing the duration of delay before the CGM signal results in the enunciation of a high threshold alarm.
(20) In another aspect, the present invention a method of determining a latest insulin delivery amount and a temporary insulin delivery rate to be delivered when a closed loop insulin delivery system terminates unexpectedly, comprising a) providing a glucose level and predicting a future glucose level in order to determine the appropriate insulin amount for the latest control action, and assuming that this commanded is immediately followed by open loop control where the preprogrammed temporary basal rate will be in effect thereafter; b) analyzing the future glucose level using this latest insulin amount plus a future temporary basal rate to determine if the future glucose level is acceptable, and if so, waiting for a selected period of time and repeating a); c) if the

glucose level of b) is not acceptable, computing an alternate maximum temporary basal delivery amount that is lower than the previously assumed basal rate, such that the predicted future glucose level is acceptable, and if so, providing a temporary basal command at the computed basal rate and duration, and predicting a future glucose level; and if so, waiting for a selected period of time and repeating a); d) if the predicted future glucose level is unacceptable, projecting a future glucose level using the lowest temporary basal delivery rate computable in c) and a maximum duration that is shorter than the maximum delivery duration, and, if the future glucose level is acceptable, providing a temporary basal command including the rate and duration of along with a nominal latest insulin delivery command and then wait for a selected period of time and repeat a); and f) reducing the latest insulin delivery command by a predetermined minimum insulin delivery resolution and repeat a) such that the combination of the reduced latest insulin delivery amount and a suitable temporary basal rate results in an acceptable predicted glucose profile if calculating a lower alternate temporary basal rate or a lower and shorter temporary basal rate does not result in an acceptable future glucose level.
(21) Other features and advantages of the invention will become apparent from the following detailed description, taken in conjunction with the accompanying drawings, which illustrate, by way of example, the features of the invention.

Description

BRIEF DESCRIPTION OF THE DRAWINGS
(1) FIG. **1** is a schematic diagram illustrating an exemplary embodiment of a controller and its various components in operable communication with one or more medical devices, such as a glucose monitor or drug delivery pump, and optionally, in operable communication with a remote controller device.
(2) FIG. **2** is a graphical representation of a glucose profile showing glucose level measured using a CGM sensor as a function of time, and also showing the variation of the glucose level as function of carbohydrate intake and insulin administration.
(3) FIG. **3** is a block diagram illustrating an embodiment of the present invention.
DETAILED DESCRIPTION OF THE PREFERRED EMBODIMENTS
(4) For the purposes of promoting an understanding of the principles of the invention, reference will now be made to a number of illustrative embodiments shown in the attached drawings and specific language will be used to describe the same. It will be understood that throughout this document, the terms “user” and “patient” are used interchangeably.
(5) Referring now to FIG. **1**, a block diagram of one illustrative embodiment of a system **10** for determining drug administration information is shown. In the illustrated embodiment, the system **10** includes an electronic device **12**, which may be handheld, having a processor **14** in data communication with a memory unit **16**, an input device **18**, a display **20**, and a communication input/output unit **24**. The electronic device **12** may be provided in the form of a general purpose computer, central server, personal computer (PC), lap top or notebook computer, personal data assistant (PDA), programmable telephone or cellular phone or other hand-held device, external infusion pump, glucose meter, analyte sensing system, or the like. The electronic device **12** may be configured to operate in accordance with one or more conventional operating systems including for example, but not limited to, the Windows® operating system (distributed by Microsoft Corporation), the Linux operating system, the Mac OS® (distributed by Apple, Inc.) and embedded operating systems such as the QNX® operating system (distributed by QNX Software Systems), the eCOS® operating system (distributed by eCosCentric Limited), Windows CE® (distributed by Microsoft Corporation) and the Palm® operating system (distributed by Palm Inc.), and may be configured to process data according to one or more conventional internet protocols for example,

but not limited to, NetBios, TCP/IP and AppleTalk® (Apple, Inc.). In any case, the electronic device **12** forms part of a fully closed-loop, semi closed-loop, or open loop diabetes control system.
(6) The processor **14** is microprocessor-based, although processor **14** may alternatively be formed of one or more general purpose and/or application specific circuits and operable as described hereinafter. The processor **14** is programmed using appropriate software commands that may be stored in the memory or communicated to the processor **14** as needed. The memory unit **16** includes sufficient capacity to store data, one or more software algorithms executable by the processor **14** and other data. The memory unit **16** may include one or more conventional memory or other data storage devices. Electronic device **12** may also include an integrated glucose meter for use in calibrating a continuous glucose monitor (CGM) or for calculating insulin amounts for bolus delivery.
(7) The input device **18** may be used in a conventional manner to input and/or modify data. The display **20** is also included for viewing information relating to operation of the device **12** and/or system **10**. Such a display may be a conventional display device including for example, but not limited to, a light emitting diode (LED) display, a liquid crystal display (LCD), a cathode ray tube (CRT) display, or the like. Alternatively or additionally, the display **20** may be or include an audible display configured to communicate information to a user, another person, or another electronic system having audio recognition capabilities via one or more coded patterns, vibrations, synthesized voice responses, or the like. Alternatively or additionally, the display **20** may be or include one or more tactile indicators configured to display or annunciate tactile information that may be discerned by the user or another person.
(8) The input device **18** may be or include a conventional keyboard or keypad for entering alphanumeric data into the processor **14**. Such a keyboard or keypad may include one or more keys or buttons configured with one or more tactile indicators to allow users with poor eyesight to find and select an appropriate one or more of the keys, and/or to allow users to find and select an appropriate one or more of the keys in poor lighting conditions. Alternatively or additionally, the input device **18** may be or include a conventional mouse or other conventional point and click device for selecting information presented on the display **20**. Alternatively or additionally, the input device **18** may include the display **20** configured as a graphical user interface (GUI). In this embodiment, the display **20** may include one or more selectable inputs that a user may select by touching an appropriate portion of the display **20** using an appropriate implement. Alternatively, the display **20** may be configured as a touch-screen capable of responding to user activation to, for example, enter data or select device functions.
(9) Alternatively, the input device **18** may also include a number of switches or buttons that may be activated by a user to select corresponding operational features of the device **12** and/or system **10**. Input device **18** may also be or include voice-activated circuitry responsive to voice commands to provide corresponding input data to the processor **14**. In any case, the input device **18** and/or display **20** may be included with or separate from the electronic device **12**.
(10) System **10** may also include a number of medical devices which carry out various functions, for example, but not limited to, monitoring, sensing, diagnostic, communication and treatment functions. In such embodiments, any of the one or more of the medical devices may be implanted within the user's body, coupled externally to the user's body (e.g., such as an infusion pump), or separate from the user's body. Alternatively or additionally, one or more of the medical devices may be mounted to and/or form part of the electronic device **12**. Typically, the medical devices are each configured to communicate wirelessly with the communication I/O unit **24** of the electronic device **12** via one of a corresponding number of wireless communication links.
(11) The wireless communications between the various components of the system **10** may be one-way or two-way. The form of wireless communication used may include, but is not limited to, radio frequency (RF) communication, infrared (IR) communication, Wi-Fi, RFID (inductive coupling) communication, acoustic communication, capacitive signaling (through a conductive body),

galvanic signaling (through a conductive body), or the like. In any such case, the electronic device **12** and each of the medical devices include conventional circuitry for conducting such wireless communications circuit. Alternatively, one or more of the medical devices may be configured to communicate with the electronic device **12** via one or more conventional serial or parallel configured hardwire connections therebetween.
(12) Each of the one or more medical devices **26** may include one or more of a conventional processing unit **52**, conventional input/output circuitry and/or devices **56**, **58** communication ports **60** and one or more suitable data and/or program storage devices **58**. It will be understood that not all medical devices **26** will have the same componentry, but rather will only have the components necessary to carry out the designed function of the medical device. For example, in one embodiment, a medical device **26** may be capable of integration with electronic device **12** and remote device **30**. In another embodiment, medical device may also be capable of standalone operation, should communication with electronic device **12** or remote device **30** be interrupted. In another embodiment, medical device **26** may include processor, memory and communication capability, but does not have a display **58** or input **56**. In still another embodiment, the medical device **26** may include an input **56**, but lack a display **58**.
(13) In some embodiments, the system **10** may alternatively or additionally include a remote device **30**. The remote device **30** may include a processor **32**, which may be identical or similar to the processor **14**, a memory or other data storage unit **34**, an input device **36**, which may be or include any one or more of the input devices described hereinabove with respect to the input device **18**, a display unit **38**, which may be or include any one or more of the display units described hereinabove with respect to the display unit **20**, and communication I/O circuitry **40**. The remote device **30** may be configured to communicate with the electronic device **12** or medical devices(s) **26** via any wired or wireless communication interface **42**, which may be or include any of the communication interfaces or links described hereinabove. Although not shown, remote device **30** may also be configured to communicate directly with one or more medical devices **26**, instead of communicating with the medical device **26** through electronic device **12**.
(14) The system **10** illustrated in FIG. **1** is, or forms part of, a fully closed-loop, semi closed-loop, or open loop diabetes control arrangement. In this regard, the system **10** requires user input of some amount of information from which the system **10** determines, at least in part, insulin bolus administration information. Such insulin bolus administration information may be or include, for example, insulin bolus quantity or quantities, bolus type, insulin bolus delivery time, times or intervals (e.g., single delivery, multiple discrete deliveries, continuous delivery, etc.), and the like. Examples of user supplied information may be, for example but not limited to, user glucose concentration, interstitial glucose level information, information relating to a meal or snack that has been ingested, is being ingested, or is to be ingested sometime in the future, user exercise information, user stress information, user illness information, information relating to the user's menstrual cycle, and the like. In any case, the system **10** includes a delivery mechanism for delivering controlled amounts of a drug; such as, for example, insulin, glucagon, incretin, or the like, and/or offering an alternatively actionable therapy recommendation to the user via the display **20**, such as, for example, directions or instructions related to ingesting carbohydrates, exercising, and the like.
(15) The system **10** may be provided in any of a variety of configurations, and examples of some such configurations will now be described. It will be understood, however, that the following examples are provided merely for illustrative purposes, and should not be considered limiting in any way. Those skilled in the art may recognize other possible implementations of a fully closed-loop, semi closed-loop, or open loop diabetes control arrangement, and any such other implementations are contemplated by this disclosure.
(16) In a first exemplary implementation of the system **10**, the electronic device **12** is provided in the form of an insulin pump configured to be worn externally to the user's body and also configured

to controllably deliver insulin to the user's body. In this example, the medical devices **26** may include one or more implanted sensors and/or sensor techniques for providing information relating to the physiological condition of the user. Examples of such implanted sensors may include, but should not be limited to, a glucose sensor, a body temperature sensor, a blood pressure sensor, a heart rate sensor, one or more bio-markers configured to capture one or more physiological states of the body, such as, for example, HBA1C, or the like.

(17) In implementations that include an implanted glucose sensor, the system **10** may be a fully closed-loop system operable in a conventional manner to automatically monitor glucose and deliver insulin, as appropriate, to maintain glucose at desired levels. The various medical devices may alternatively or additionally include one or more sensors or sensing systems that are external to the user's body and/or sensor techniques for providing information relating to the physiological condition of the user. Examples of such sensors or sensing systems may include, but should not be limited to, a glucose strip sensor/meter, a body temperature sensor, a blood pressure sensor, a heart rate sensor, one or more bio-markers configured to capture one or more physiological states of the body, such as, for example, HBA1C, or the like.

(18) In implementations that include an external glucose sensor, the system **10** may be a closed-loop, semi closed-loop, or open loop system operable in a conventional manner to deliver insulin, as appropriate, based on glucose information provided thereto by the user. Information provided by any such sensors and/or sensor techniques may be communicated to the system **10** using any one or more conventional wired or wireless communication techniques. In this exemplary implementation, the remote device **30** may also be included in the form of a handheld or otherwise portable electronic device configured to communicate information to and/or from the electronic device **12**.

(19) In a second exemplary implementation of the system **10**, the electronic device **12** is provided in the form of a handheld remote device, such as a PDA, programmable cellular phone, or other handheld device. In this example, the medical devices **26** include at least one conventional implantable or externally worn drug pump. In one embodiment of this example, an insulin pump is configured to controllably deliver insulin to the user's body. In this embodiment, the insulin pump is configured to wirelessly transmit information relating to insulin delivery to the handheld device **12**. The handheld device **12** is configured to monitor insulin delivery by the pump, and may further be configured to determine and recommend insulin bolus amounts, carbohydrate intake, exercise, and the like. The system **10** may or may not be configured in this embodiment to provide for transmission of wireless information from the handheld device **12** to the insulin pump.

(20) In an alternate embodiment of this example, the handheld device **12** is configured to control insulin delivery to the user by determining insulin delivery commands and transmitting such commands to the insulin pump. The insulin pump, in turn, is configured to receive the insulin delivery commands from the handheld device **12**, and to deliver insulin to the user according to the commands. The insulin pump, in this embodiment, may or may not further process the insulin pump commands provided by the handheld unit **12**. In any case, the system **10** will typically be configured in this embodiment to provide for transmission of wireless information from the insulin pump back to the handheld device **12** to thereby allow for monitoring of pump operation. In either embodiment of this example, the system **10** may further include one or more implanted and/or external sensors of the type described in the previous example. In this exemplary implementation, a remote device **30** may also be included in the form of, for example, a PC, PDA, programmable cellular phone, laptop or notebook computer configured to communicate information to and/or from the electronic device **12**.

(21) Those skilled in the art will recognize other possible implementations of a fully closed-loop, semi closed-loop, or open loop diabetes control arrangement using at least some of the components of the system **10** illustrated in FIG. **1**. For example, the electronic device **12** in one or more of the above examples may be provided in the form of a PDA, programmable cellular phone, laptop, notebook or personal computer configured to communicate with one or more of the medical

devices **26**, at least one of which is an insulin delivery system, to monitor and/or control the delivery of insulin to the user. As another example, the remote device **30** may be configured to communicate with the electronic device **12** and/or one or more of the medical devices **26**, to control and/or monitor insulin delivery to the patient, and/or to transfer one or more software programs and/or data to the electronic device **12**. The remote device **30** may reside in a caregiver's office or other remote location, and communication between the remote device and any component of the system **10** may be accomplished via an intranet, internet (such as, for example, through the world-wide-web), cellular, telephone modem, RF, or other communication link. Any one or more conventional internet protocols may be used in such communications. Alternatively or additionally, any conventional mobile content delivery system; such as, for example, Wi-Fi, WiMAX, short message system (SMS), or other conventional message scheme may be used to provide for communication between devices comprising the system **10**.
(22) Generally, the concentration of glucose in a person changes as a result of one or more external influences such as meals and exercise, and also changes resulting from various physiological mechanisms such as stress, illness, menstrual cycle and the like. In a person with diabetes, such changes can necessitate monitoring the person's glucose level and administering insulin or other glucose level altering drug, such as, for example, a glucose lowering or raising drug, as needed to maintain the person's glucose level within a desired range. In any of the above examples, the system **10** is thus configured to determine, based on some amount of patient-specific information, an appropriate amount, type and/or timing of insulin or other glucose level altering drug to administer in order to maintain normal glucose levels without causing hypoglycemia or hyperglycemia. In some embodiments, the system **10** is configured to control one or more external insulin pumps, such as, for example, subcutaneous, transcutaneous or transdermal pumps, and/or implanted insulin pumps to automatically infuse or otherwise supply the appropriate amount and type of insulin to the user's body in the form of one or more insulin boluses.
(23) In other embodiments, the system **10** is configured to display or otherwise notify the user of the appropriate amount, type, and/or timing of insulin in the form of an insulin delivery or administration recommendation or instruction. In such embodiments, the hardware and/or software forming system **10** allows the user to accept the recommended insulin amount, type, and/or timing, or to reject it. If the recommendation is accepted by the user, the system **10**, in one embodiment, automatically infuses or otherwise provides the appropriate amount and type of insulin to the user's body in the form of one or more insulin boluses. If, on the other hand, the user rejects the insulin recommendation, the hardware and/or software forming system **10** allows the user to override the system **10** and manually enter values for insulin bolus quantity, type, and/or timing in the system. The system **10** is thus configured by the user to automatically infuse or otherwise provide the user specified amount, type, and/or timing of insulin to the user's body in the form of one or more insulin boluses.
(24) Alternatively, the appropriate amount and type of insulin corresponding to the insulin recommendation displayed by the system **10** may be manually injected into, or otherwise administered to, the user's body. It will be understood, however, that the system **10** may alternatively or additionally be configured in like manner to determine, recommend, and/or deliver other types of medication to a patient.
(25) The system **10** is operable, as just described, to determine and either recommend or administer an appropriate amount of insulin or other glucose level lowering drug to the patient in the form of one or more insulin boluses. In order to determine appropriate amounts of insulin to be delivered or administered to the user to bring the user's glucose level within an acceptable range, the system **10** requires at least some information relating to one or more external influences and/or various physiological mechanisms associated with the user. For example, if the user is about to ingest, is ingesting, or has recently ingested, a meal or snack, the system **10** generally requires some information relating to the meal or snack to determine an appropriate amount, type and/or timing of

one or more meal compensation boluses of insulin. When a person ingests food in the form of a meal or snack, the person's body reacts by absorbing glucose from the meal or snack over time. For purposes of this document, any ingesting of food may be referred to hereinafter as a “meal,” and the term “meal” therefore encompasses traditional meals, such as, for example, breakfast, lunch and dinner, as well as intermediate snacks, drinks, and the like.

(26) FIG. **2** depicts a typical glucose absorption profile **200** for a user measured using a CGM sensor. The graph **205** plots the measured glucose level as a function of time. This profile shows the effect on glucose level of various actions, such as carbohydrate intake **210**, and the delivery of rapid acting insulin **210** and long acting insulin **230**.

(27) The general shape of a glucose absorption profile for any person rises following ingestion of the meal, peaks at some measurable time following the meal, and then decreases thereafter. The speed, that is, the rate from beginning to completion, of any one glucose absorption profile typically varies for a person by meal composition, meal type or time (such as, for example, breakfast, lunch, dinner, or snack) and/or according to one or more other factors, and may also vary from day-to-day under otherwise identical meal circumstances. Generally, the information relating to such meal intake information supplied by the user to the system **10** should contain, either explicitly or implicitly, an estimate of the carbohydrate content of the meal or snack, corresponding to the amount of carbohydrates that the user is about to ingest, is ingesting, or has recently ingested, as well as an estimate of the speed of overall glucose absorption from the meal by the user.

(28) The estimate of the amount of carbohydrates that the patient is about to ingest, is ingesting, or has recently ingested, may be provided by the user in any of various forms. Examples include, but are not limited to, a direct estimate of carbohydrate weight (such as, for example, in units of grams or other convenient weight measure), an amount of carbohydrates relative to a reference amount (such as, for example, dimensionless), an estimate of meal or snack size (such as, for example, dimensionless), and an estimate of meal or snack size relative to a reference meal or snack size (such as, for example, dimensionless). Other forms of providing for user input of carbohydrate content of a meal or snack will occur to those skilled in the art, and any such other forms are contemplated by this disclosure.

(29) The estimate of the speed of overall glucose absorption from the meal by the user may likewise be provided by the user in any of various forms. For example, for a specified value of the expected speed of overall glucose absorption, the glucose absorption profile captures the speed of absorption of the meal taken by the user. As another example, the speed of overall glucose absorption from the meal by the user also includes time duration between ingesting of the meal by a user and the peak glucose absorption of the meal by that user, which captures the duration of the meal taken by the user. The speed of overall glucose absorption may thus be expressed in the form of meal speed or duration. Examples of the expected speed of overall glucose absorption parameter in this case may include, but are not limited to, a compound parameter corresponding to an estimate of the meal speed or duration (such as, for example, units of time), a compound parameter corresponding to meal speed or duration relative to a reference meal speed or duration (such as, for example, dimensionless), or the like.

(30) As another example of providing the estimate of the expected speed of overall glucose absorption parameter, the shape and duration of the glucose absorption profile may be mapped to the composition of the meal. Examples of the expected speed of overall glucose absorption parameter in this case may include, but are not limited to, an estimate of fat amount, protein amount and carbohydrate amount (such as, for example, in units of grams) in conjunction with a carbohydrate content estimate in the form of meal size or relative meal size, an estimate of fat amount, protein amount and carbohydrate amount relative to reference fat, protein and carbohydrate amounts in conjunction with a carbohydrate content estimate in the form of meal size or relative meal size, and an estimate of a total glycemic index of the meal or snack (such as, for

example, dimensionless), wherein the term “total glycemic index” is defined for purposes of this document as a parameter that ranks meals and snacks by the speed at which the meals or snacks cause the user's glucose level to rise. Thus, for example, a meal or snack having a low glycemic index produces a gradual rise in glucose level whereas a meal or snack having a high glycemic index produces a fast rise in glucose level. One exemplary measure of total glycemic index may be, but is not limited to, the ratio of carbohydrates absorbed from the meal and a reference value, such as, for example, derived from pure sugar or white bread, over a specified time period, such as, for example, 2 hours. Other forms of providing for user input of the expected overall speed of glucose absorption from the meal by the patient, and/or for providing for user input of the expected shape and duration of the glucose absorption profile generally will occur to those skilled in the art, and any such other forms are contemplated by this disclosure. [**0054**] Generally, the concentration of glucose in a person with diabetes changes as a result of one or more external influences such as meals and/or exercise, and may also change resulting from various physiological mechanisms such as stress, menstrual cycle and/or illness. In any of the above examples, the system **10** responds to the measured glucose by determining the appropriate amount of insulin to administer in order to maintain normal glucose levels without causing hypoglycemia. In some embodiments, the system **10** is implemented as a discrete system with an appropriate sampling rate, which may be periodic, aperiodic or triggered, although other continuous systems or hybrid systems may alternatively be implemented as described above.

(31) As one example of a conventional diabetes control system, one or more software algorithms may include a collection of rule sets which use (1) glucose information, (2) insulin delivery information, and/or (3) user inputs such as meal intake, exercise, stress, illness and/or other physiological properties to provide therapy, and the like, to manage the user's glucose level. The rule sets are generally based on observations and clinical practices as well as mathematical models derived through or based on analysis of physiological mechanisms obtained from clinical studies. In the exemplary system, models of insulin pharmacokinetics and pharmacodynamics, glucose pharmacodynamics, meal absorption and exercise responses of individual patients are used to determine the timing and the amount of insulin to be delivered. A learning module may be provided to allow adjustment of the model parameters when the patient's overall performance metric degrades such as, for example, adaptive algorithms, using Bayesian estimates, may be implemented. An analysis model may also be incorporated which oversees the learning to accept or reject learning. Adjustments are achieved utilizing heuristics, rules, formulae, minimization of cost function(s) or tables (such as, for example, gain scheduling).

(32) Predictive models can be programmed into the processor(s) of the system using appropriate embedded or inputted software to predict the outcome of adding a controlled amount of insulin or other drug to a user in terms of the an expected glucose value. The structures and parameters of the models define the anticipated behavior.

(33) Any of a variety of conventional controller design methodologies, such as PID systems, full state feedback systems with state estimators, output feedback systems, LQG (Linear-Quadratic-Gaussian) controllers, LQR (Linear-Quadratic-Regulator) controllers, eigenvalue/eigenstructure controller systems, and the like, could be used to design algorithms to perform physiological control. They typically function by using information derived from physiological measurements and/or user inputs to determine the appropriate control action to use. While the simpler forms of such controllers use fixed parameters (and therefore rules) for computing the magnitude of control action, the parameters in more sophisticated forms of such controllers may use one or more dynamic parameters. The one or more dynamic parameters could, for example, take the form of one or more continuously or discretely adjustable gain values. Specific rules for adjusting such gains could, for example, be defined either on an individual basis or on the basis of a user population, and in either case will typically be derived according to one or more mathematical models. Such gains are typically scheduled according to one or more rule sets designed to cover the expected

operating ranges in which operation is typically nonlinear and variable, thereby reducing sources of error.
(34) Model based control systems, such as those utilizing model predictive control algorithms, can be constructed as a black box wherein equations and parameters have no strict analogs in physiology. Rather, such models may instead be representations that are adequate for the purpose of physiological control. The parameters are typically determined from measurements of physiological parameters such as glucose level, insulin concentration, and the like, and from physiological inputs such as food intake, alcohol intake, insulin doses, and the like, and also from physiological states such as stress level, exercise intensity and duration, menstrual cycle phase, and the like. These models are used to estimate current glucose level or to predict future glucose levels. Such models may also take into account unused insulin remaining in the user after a bolus of insulin is given, for example, in anticipation of a meal. Such unused insulin will be variously described as unused, remaining, or “insulin on board.”
(35) Insulin therapy is derived by the system based on the model's ability to predict glucose levels for various inputs. Other conventional modeling techniques may be additionally or alternatively used to predict glucose levels, including for example, but not limited to, building models from first principles.
(36) In a system as described above, the controller is typically programmed to provide a “basal rate” of insulin delivery or administration. Such a basal rate is the rate of continuous supply of insulin by an insulin delivery device such as a pump that is used to maintain a desired glucose level in the user. Periodically, due to various events that affect the metabolism of a user, such as eating a meal or engaging in exercise, a “bolus” delivery of insulin is required. A “bolus” is defined as a specific amount of insulin that is required to raise the blood concentration of insulin to an effective level to counteract the affects of the ingestion of carbohydrates during a meal and also takes into account the affects of exercise on the glucose level of the user.
(37) As described above, an analyte monitor may be used to continuously monitor the glucose level of a user. The controller is programmed with appropriate software and uses models as described above to predict the effect of carbohydrate ingestion and exercise, among other factors, on the predicted level of glucose of the user at a selected time. Such a model must also take into account the amount of insulin remaining in the blood stream from a previous bolus or basal rate infusion of insulin when determining whether or not to provide a bolus of insulin to the user.
(38) In a typical situation, an insulin pump normally delivers insulin without user intervention when delivering the insulin at the basal insulin delivery rate. At times, however, the pump may detect conditions that warrant providing an alarm or other signal to the user that intervention in the insulin delivery by the user is necessary. Depending on the conditions and the user's response, the controller may instruct the pump to stop delivering insulin.
(39) When insulin delivery is stopped, the user will eventually need to restart insulin delivery. In view of the period of time during which insulin delivery was stopped, the user may need to deliver a correction bolus of insulin to bring their glucose level back within an acceptable range. The volume of the insulin bolus is calculated by the user from their current glucose level and their experience with managing their diabetes. For example, users who have experience in managing their diabetes are typically able to estimate the various factors that need to be considered in calculating a corrective bolus dosage of insulin.
(40) In one example, the size of the corrective bolus required to maintain the user in euglycemia is related to the manner and duration in which delivery of insulin was stopped. For example, if insulin delivery was stopped for a few minutes, perhaps no corrective bolus of insulin is required. However, if insulin delivery is stopped for several hours, a significant bolus of insulin may be required to bring the user's glucose level into an acceptable range.
(41) The controller of the system monitors the insulin pump and when the pump determines that the user wants to restart the pump, the pump may either confirm that the user wants to restart basal rate

delivery of insulin, or it may prompt the user to administer a manual bolus of insulin.
(42) In one embodiment of the invention, the controller and/or pump has a memory that stores information related to the history of the user's glucose levels and various actions or events that have been taken to adjust those levels, such as the rate of basal delivery of insulin, the amount of the last insulin bolus delivery, and the time between various events or user actions. The controller and/or pump may also store the time that has elapsed since the pump was stopped. Additional factors that may be considered include the user's insulin sensitivity, the amount of insulin that may remain in the user's blood stream since the last basal or bolus insulin delivery and whether or not an event or action may have incurred that would cause the basal rate of insulin delivery to have changed since the pump was stopped.
(43) When the user desires to restart the pump, the controller and/or pump may, using the data programmed within the memory regarding the user's history of insulin delivery and glucose levels resulting therefrom, calculate a bolus volume of insulin to deliver using a model such as set forth as:
Remaining insulin=delivered volume×(IATIME−time)/IATIME,
(44) Where: Delivered volume is the amount of insulin delivered, time is the time since the insulin was delivered, and IATIME is the user's insulin action time, which is a measure of how long insulin remains active in an individual user's body.
(45) Those skilled in the art will understand that other models can be used to perform this estimation, such as, for example, the decaying exponential model or model using an S-model decay.
(46) For basal delivery, the times since the insulin was last delivered, or the age of the delivery, is not a constant. The calculation is a convolution of the basal profile function and the linear decay function.
(47) In practice, basal insulin delivery is not delivered continuously, but instead is accumulated and delivered at discrete times. For example, an exemplary insulin pump may be programmed to deliver three minutes of basal rate insulin delivery spaced three minutes apart. This simplifies the above calculation to the dot product of two vectors. For example, D=vector of deliveries that were not performed, oldest to newest. For example, D=(0.3, 0.3, 0.3, 0.3, 0.3, 0.3, 0.3, 0.3, 0.3, 0.3, 0.5, 0.5, 0.5, 0.5, 0.5, 0.5, 0.5, 0.5, 0.5, 0.5) which represents 20 deliveries of 0.3 and 0.5 units spaced three minutes apart. Similarly, T=vector or time multipliers. For example, if the insulin action time of the patient is 60 minutes and delivery has been stopped for 42 minutes, then: T equals (0.00, 0.00, 0.00, 0.00, 0.00, 0.00, 0.35, 0.40, 0.45, 0.50, 0.55, 0.60, 0.65, 0.70, 0.75, 0.80, 0.85, 0.90, 0.95, 1.00). The dot product of D*T equal 4.39 units.
(48) Thus, for this example, the user will need to either manually administer or program the pump to administer 4.39 units of insulin, the bolus volume that will provide the same value for the insulin onboard for the user had the basal delivery of insulin not been stopped.
(49) In another embodiment of the present invention using a closed looped control system, the controller or pump may be programmed to perform a model-based calculation of transitory insulin using an insulin bolus command as the nominal input for the model. This process provides a means to use model-based calculation of insulin onboard (JOB) past insulin delivery information, and past and present glucose information and predicted glucose profile over a near-future horizon to determine a safe amount and duration of basal rate in the event that a closed loop system that normally uses bolus insulin deliver terminates its operation unexpectedly.
(50) Automated closed loop systems that regulate glucose levels by using insulin delivery have to account for instances when the closed loop system terminates and switches to open loop control. For users with insulin pumps, open-loop operation typically includes a pre-programmed insulin basal rate as well as a bolus insulin delivery rate administered in response to known events, such as a meal. When a system transitions from open to closed loop, past insulin delivery information is available from the memory and can be used by the closed loop controller to determine the best and

safest insulin delivery profile for the user.
(51) One hazard mitigated by this process is related to the transition from closed loop to open loop operation, such as, for example, in the case where a controller under closed loop control has determined a series of insulin commands that may require subsequent insulin deliveries to be lower than the open loop pre-programmed basal insulin delivery rate. As long as the system remains in closed loop operation, this is not an issue. However, should the system transition to open loop operation, having the pump deliver insulin at the pre-programmed basal rate could result in unwanted hypoglycemia if the controller previously ordered the pump to deliver insulin at a rate and duration that resulted in a large residual IOB with a plan to control the delivery in the future to significantly reduce the insulin delivery at a rate below the pre-programmed basal rate.
(52) One embodiment of the invention addresses the situation where the closed loop system operates by delivering bolus insulin commands at a fixed interval in time, and the closed-loop system periodically obtains glucose measurement from a glucose measuring device. Such a system is also capable of issuing a temporary basal rate command to the pump with a specified amount and duration in which the basal command should apply since the command was issued. When the duration of such a demand expires, the system reverts to the pre-programmed basal rate.
(53) As used in this description of the various embodiments of the invention, the latest command intended for normal operation of the system is described in terms of “bolus commands.” Alternatively, the safety command intended to be used when closed loop operation of the system is interrupted is described in terms of a “temporary basal rate” command. The replacement of the normal operation command in terms of bolus with basal rate does not change the mechanism of the various embodiments of the invention. Using bolus and basal rate for the normal operation and the safety-related commands clarifies the roles of those commands in the description of the various embodiments of the invention included herein.
(54) In another embodiment of the invention, the closed loop system is able to track the IOB amount by utilizing insulin pharmacokinetic and pharmacodynamic models, and by keeping track of insulin delivery data corresponding to the actual delivery of insulin to the user. This delivery data includes bolus and basal insulin amounts commanded by the user or by the automated closed loop controller.
(55) For example, the model used by the controller to calculate insulin delivery rates may utilize a pharmacokinetic model, such as a modified version of a model used by Hovorka et al.:
(56) $\frac{\partial(i1(t))}{\partial t} = \frac{-1(t)}{tau\_i} + u(t)\frac{\partial(i2(t))}{\partial(t)} = -\frac{[i2(t) - i1(t)]}{tau\_i} i(t) = \frac{i2(t)}{MCRWtau\_i}$
(57) Where: (t) denotes variables and/or states that vary in time; u(t) is the amount of insulin bolus given at any time t; i1(t) and i2(t) are internal states of the insulin absorption; i(t) is the plasma insulin concentration; tau i is the time-to-peak of insulin absorption; MCR is the metabolic clearance rate; and W is the user's weight.
(58) An example of an insulin pharmacodynamic model that may be programmed into the controller or pump using suitable software or hardware commands is given by:
(59)
$\frac{\partial(x(t))}{\partial t} = -p1[x()t) - i(t)]\frac{\partial(g(t))}{dt} = -[Sg + [Si .Math. x(t)]]g(t) = [Sg .Math. g)(t)] + Sm .Math. \frac{\partial(gm(t))}{dt}]$
(60) Where: Sg is the glucose clearance rated independent of insulin; Si is the glucose clearance rate due to insulin; g0(t) is a slowly varying equilibrium for the glucose clearance independent of insulin; Sm is the glucose appearance constant due to meal absorption; and
(61) $\frac{\partial(gm(t)}{\partial t}$
is the rate of glucose appearance due to meal absorption.
(62) In this model, the first equation governs the effective insulin x(t) over time as a function of plasma insulin i(t), given a decay time constant of 1/pl. The second equation governs how glucose g(t) varies over time as a function of many factors, including, for example, the insulin pharmacodynamics −Si x(t) g(t).

(63) At any given time, the closed loop system is able to estimate the latest glucose by utilizing the IOB data, insulin history, glucose measurements, and a physiological model of glucose in response to insulin and other measurable factors, such as exercise and meal estimate and/or announcements. Additionally, at any time, the closed loop system in one embodiment of the invention is able to compute a desired bolus command that would result in the present and projected glucose of the user in a selected near future horizon falling within a specified nominal target, or one that optimizes the present and projected glucose with respect to some robust optimality criteria. This is a typical process involved in automatic closed looped systems that are model-based, such as a model-based approach that falls under the model predictive control (MPC) architecture.
(64) When the system receives a temporary basal command, the closed loop system may, given the JOB, insulin history, glucose history, predicted glucose (in the near future horizon), and pre-programmed basal insulin delivery rate, compute a temporary basal amount and duration such that if the closed loop operation of the system terminates, the combination of the basal rate of that amount and the specified duration and the subsequent transition to the pre-programmed basal rate is such that the predicted glucose value of the user in the foreseeable future horizon will remain within a specified safe target. This safe target may be different from the specified nominal target value for the optimal calculation, in that it can be designed to be more conservative from a hazards prevention perspective.
(65) An example of a process for determining a temporary basal rate of insulin delivery, along with the duration of that basal rate delivery, is illustrated by the diagram of FIG. **3**. Appropriate software or hardware commands may be used to program the processor of the pump and/or controller to carry out the illustrated steps. Using a heuristic process, a glucose value may be predicted into the near future horizon under an assumption that the commanded bolus insulin delivery at present is followed by an immediate transfer to the programmed basal rate. In other words, the temporary basal rate candidate value calculated using the heuristic by the controller is identical to the pre-programmed basal rate.
(66) Using this information as a starting point in box **310** as input into the selected model, a predicted glucose profile for a future time is calculated in box **320**. If an acceptable predicted glucose profile results from this calculation in box **330**, when there is no need to issue a temporary basal rate of any duration to maintain the user's glucose level within an acceptable range. When this is the case, the system waits until the next time an insulin command needs to be issued in box **340**. When the next insulin command is determined to be needed, the processor begins at box **310**, and once again carries out the process wherein glucose is predicted into the near future horizon as stated above.
(67) If an acceptable predicted glucose profile does not result from the calculation carried out by the processor of the controller, the temporary basal insulin delivery rate is decreased by one value lower from the previous temporary basal rate candidate in box **350**. The resolution of the basal rate for the particular pump and/or controller and/or glucose sensor determines the size of the reduction of basal rate that will be applied. This calculation may assume the maximum delivery duration allowable by the system.
(68) If this assumption results in an acceptable predicted glucose profile in box **360**, then a temporary basal insulin delivery command is sent using this basal rate and duration along with the nominal bolus command to the pump in box **370**. The controller then waits until the next time an insulin command needs to be issued, and then restarts the process at box **310** again.
(69) If an acceptable glucose profile is not calculated in box **360**, the process keeps the same temporary basal rate as was used in box **350**, and the decreases the duration of the basal rate delivery by an amount that is a function of system's resolution on command duration in box **380**. If this calculation results in an acceptable predicted glucose profile in box **290**, then a temporary basal command using this basal rate and duration along with the nominal bolus command is sent to the pump in box **370**. Again, the controller waits until the next time an insulin command needs to be

issued in box **340**. When that time occurs, the process begins again box **310**.
(70) The iterative process described above is continued until the lowest allowable basal rate of insulin delivery is reached. If this assumption does not result in an acceptable predicted glucose profile, then the bolus command is reduced such that the combination of the reduced bolus rate and the temporary basal rate results in an acceptable predicted glucose profile. If this is not possible, then an alert is sent to the user to take rescue carbohydrates or other actions that counteract the effect of excessive insulin.
(71) The purpose of the above-described process is to add an additional safety calculation layer for the process control provided by automated closed-loop algorithms such that the closed-loop system computes two scenarios. First, the common scenario where insulin commands are issued and acted upon without interruption in the foreseeable future. This is normally computed by the automated closed-loop systems using model based approaches such as MPC. Second, the hazard-mitigated scenario where the sequence of insulin commands are abruptly interrupted after the latest command. The process set forth above addresses this scenario.
(72) Using such a system, the embodiments of the present invention use model-based principles and available data to predict more than one scenario of insulin delivery such that when the periodic normal insulin command is interrupted, the system does not cause an unwanted hypoglycemic hazard. Avoiding unwanted hypoglycemia is important in providing proper control of glucose level in the diabetic user. The various embodiments of the invention provide better hazard mitigation against the risk of hypoglycemia due to a combination of insulin stacking (where a user already has an amount of insulin on board, but administers an additional bolus of insulin, often driving the user's glucose level towards hypoglycemia) and termination of closed-loop control. Utilizing a model-based approach improves the hazard mitigation caused by interruption of the automated closed-loop system, even though the interruption may have been accompanied by an alert/alarm or a temporary basal rate with a fixed duration.
(73) In a CGM device such as the FreeStyle Navigator® Continuous Glucose Monitoring System that is distributed by Abbott Diabetes Care, there is a mute alarm feature that allows the user to mute the alarm when it is not easy for the user to respond to the alarm. Such alarms occur, for example, when the user's glucose level is either in the hyperglycemic range or, more importantly, in the hypoglycemic range.
(74) For example, the user may choose to mute the alarm for the next two hours knowing that the user will be in a long meeting and does not want the alarm to disturb anyone. However, in the case of a low alarm signifying hypoglycemia, muting the alarm may lead to failure of the user to treat the hypoglycemic condition, which could be dangerous. The situation would be improved if at the time the user is muting the alarm, the system checks to determine the likelihood that the user may become hypoglycemic within the period of time that the alarm is muted. Generally, the system calculates a projected low-glucose level using IOB and real-time information to fine tune the projection and providing an alarm to indicate low glucose values. The problem with a persistent projected alarm, however, is the additional user annoyance associated with false positives. A user-initiated approach allows the user to define the time period that he wishes to have more precaution built into the control system as opposed to the system simply providing a persistent alarm to the user indicating that the controller has determined that some correction action is necessary, but at a time that a user is either unable or unwilling to take such action.
(75) In one embodiment, the invention uses an overall strategy of potential low alarm detection to preempt pump control modification. Such a determination is useful during activities or events such as where the user goes to a long meeting or attends an event such as a movie, theatre, concert, or a long drive that may create a barrier from detecting and acting on a low-glucose level alarm. Another situation where the various embodiments of the invention are advantageous is where the user goes to bed at night and wishes to know whether a low glucose level alarm may sound during the night.

(76) This embodiment of the invention includes a predictive alarm avoidance feature that is described hereafter. When the user foresees that he will be unavailable to act on an alarm in the near future, such as, for example, in the case where the user knows he will enter a meeting within the next 30 minutes, the user initiates the process of analyzing his current CGM glucose level and pump calculated IOB information to determine if the user's glucose level is likely be at in a range that the cause the device to produce an alarm during the meeting.
(77) Using a controller that is programmed with appropriate software and hardware commands to carry out the functions required to perform the predictive alarm avoidance embodiment, the user enters the time horizon moving forward that he will be unavailable to take corrective actions if an alarm is presented. This entry is similar to the entry for the number of hours to mute an alarm in the current Freestyle Navigator® Continuous Glucose Monitoring System.
(78) Based on the time horizon entered by the user, and other information found in the memory associated with the CGM and pump device, for example, but not limited to, the current CGM glucose level reading, the current IOB reading, the current bolus calculator parameters, and the current low and high reading settings, the processor calculates whether the user will be in one of three states in the future: a) a state where his glucose profile is adequate; b) in a carbohydrate deficient state, that is, a low glucose level state; and c) an insulin deficient state, that is, the user is in a hyperglycemic state during the time that he will not be able to make an adjustment to his pump. The following example illustrates this process.
Example 1
(79) Assuming a user wishes to avoid an alarm where, and at the time he is commanding the controller to perform this calculation, the following values are pertinent to the calculation: target threshold: 110 mg/dL, high alarm threshold: 240 mg/dL, lower alarm threshold: 70 mg/dL; carbohydrate ratio: 1 unit of insulin for 15 grams of carb; insulin sensitivity factor (ISF): 1 unit of insulin for 50 mg/dL; and insulin action time: 5 hours.
(80) In the case where the user will be unavailable or unwilling to take corrective action if he is presented with an alarm for the next three hours, the processor calculates a glucose level profile of the user utilizing the above factors to determine whether the user's glucose profile will stay in an acceptable range, or if the profile will result in either a hypo- or hyper-glycemic state. In this case, with the above factors, the calculation reveals that the user would have an adequate glucose profile where the current CGM glucose value is 150 mg/dL and the current IOB is 1.1 units. Using methods well known by those in the art, the calculations carried out by the processor show that in five hours, the user's glucose level will decrease by 55 mg/dL (IOB*ISF), which is a glucose decline rate of 11 mg/dL per hour. Thus, during the three hours that the user will be unavailable or unwilling to take action to correct an alarm, the user's glucose level is estimated to be approximately 117 mg/dL (or, a 33 mg/dL decrease in glucose level from the user's current glucose level), which, for this user, is an acceptable glucose level.
(81) Now considering the case where the user's current CGM glucose value is 150 mg/dL and the pump history (stored in the memory associated with the pump or controller) indicates a current IOB of 3.1 units. Using these starting values, the processor calculates that in five hours the user's glucose level will decrease by 155 mg/dL, which is a glucose decline rate of 31 mg/dL per hour. Thus, if the user is unavailable or unwilling to take corrective action in the next three hours, his glucose level will fall to around 57 mg/dL (a 93 mg/dL decrease in glucose level from the user's current glucose level), which is a carbohydrate deficient state.
(82) Once the processor and controller knows that the user will be in a carbohydrate deficient state, the processor can recommend, for example, one or more of the following actions to the user using the display of the controller: a) instruct the user to eat X grams of carbohydrates now or later to avoid the low glucose level, the value of X being calculated as X=(target minus future low)/ISF*carb ratio which is equal to (110 minus 57)/50*1.5=15.9 grams of carbohydrates; or b) instruct the user to reduce the basal rate of his insulin pump by using a temporary basal rate on the

pump. The rate and duration of the temporary basal race is determined by the user using an interface on the device that can be interacted with by the user to cause the processor of the device to compute the temporary basal rate. For example, the controller may provide the user with the information he needs to program the processor of the device to reduce insulin delivery by X units per hour in the next three hours to i) stay within the target glucose level range, where X can be calculated as X=(target minus future low)/ISF/time horizon which is equal to (110 minus 57)/50/3 equals 0.354 units per hour (high reduction) or ii) maintain the user's glucose level above the low alarm threshold X=(low threshold minus future low)/ISF/time horizon equal (70 minus 57)/50/3 equals 0.087 units per hour (lower reduction); c) the user may also be given the option of increasing the time horizon of the calculations, especially if the recommended reduced insulin amount is greater than the current basal rate, in order to bring about a more conservative reduction in the basal rate. For example, instead of reducing the amount of insulin delivered within the next three hours, the user may extend the time of insulin delivery to five hours.

(83) If the user is watching a movie in a theatre, for example, the user may choose scenario a). If the user performs the calculation right before going to bed, or before taking a short nap, then he may choose scenario b) (assuming that the user's basal rate is greater than 0.087 units per hour or 0.35 per hour).

(84) In a further example, a user's initial CGM glucose level is 280 mg/dL and the pump history indicates that the user's current IOB is 3.1 units. In this case, the processor calculates that in five hours, the user's glucose level will decrease by 155 mg/dL, which is a glucose level decline rate of 31 mg/dL per hour. With this glucose level decline rate, during the next three hours when the user is unavailable or unwilling to take corrective action, the user's glucose level is estimated by the processor to be around 187 mg/dL (a decrease of 93 mg/dL in glucose level from the user's current level), which could mean that the user will be in an insulin deficient state.

(85) Given this calculation, the controller may also determine whether or not more insulin should be recommended to be taken by the user to avoid this possibly insulin deficient state this case, in five hours, based on the JOB, the user's glucose level can be estimated at 125 mg/dL, which a slightly above the user's target glucose level, so the controller and processor of the device may recommend slightly more insulin.

(86) Alternatively, the device may recommend that the user does not need to do anything now or that he could choose to give himself a little more insulin (X units) using an extended bolus feature of the pump where:

X=the (predicted glucose level after insulin action time)−(the target glucose level)/ISF which is equal to 0.3 units, and

(87) Time duration=the user entered time horizon which is equal to three hours.

(88) An embodiment of the controller of the present invention includes an appropriate User Interface (UI). This user interface is presented by the processor to the user and is formed by processor in accordance with appropriate software commands that have programmed the processor to operate accordingly. The user interface is user friendly to facilitate the user in making the appropriate pump changes. For example, the device should automatically initiate the UI process for temporary basal reduction in case b) and c) described above. Those skilled in the art will understand that the UI may take many different forms without departing from the scope of the present invention.

(89) Because of the acute affect of hypoglycemia, a user is likely to use the above described process to avoid low-alarm occurrence. However, it is conceivable that a similar strategy can be used to avoid exposure to the hazards of hyperglycemia.

(90) Not only can alarms be avoided using the various embodiments of the invention described above, but using a combined CGM and pump device, the actual state of low glucose level, or hypoglycemia, can be avoided. This further enhances the safety and usability of such CGM devices and insulin delivery pumps. Such an approach is also advantageous when compared to using a

CGM based projected alarm feature alone which can result in annoying false positives and has much more limited future time projection. The described approach is also superior to muting an alarm to avoid the nuisance of repeated alarms and is also less annoying than a persistent projected low alarm. The user initiated, on-demand nature of the design and the pump control modification UI process integration affords better user control and allows more proactive user intervention to reduce glycemic variability. The embodiments of the invention described above are also advantageous because they can be used as a separate bedtime basal analysis to adjust the basal insulin rate slightly before going to bed. Such an adjustment is especially useful if the user if in a carbohydrate deficient state and does not wish to eat any food before going to bed, thus giving the user the option of reducing the basal rate of insulin delivery before going to bed.

(91) Low glucose or hypoglycemia alarms are typically incorporated into continuous glucose monitoring systems to alert the user to potentially dangerous low glucose levels. For example, the low glucose alarm in the Freestyle Navigator® Continuous Glucose Monitoring System, distributed by Abbott Diabetes Care, is sounded when a user's glucose level falls below a user-defined low glucose threshold. This type of alarm may be subject to false triggering due to sensor signal anomalies such as sensor drop-outs, which are temporary and fully-recoverable signal attenuations believed to be associated with the application of pressure on the percutaneous sensor and/or transmitter as may occur during sleep.

(92) The frequency of these false low-glucose alarms may be increased during closed-loop insulin delivery as the control algorithm will attempt to normalize glucose levels, increasing the percentage of time spent in or near the euglycemic range. Sensor drop outs that occur during hyperglycemia are less likely to trigger low glucose alarms because of these elevated glucose levels. In such a case, sensor drop outs may not sufficiently attenuate the sensor signal to elicit this response. However, if a user's glucose level is maintained closer to the euglycemic (acceptable, that is, neither hypo- nor hyperglycemic) range, as is expected during close-loop control, there is an increased likelihood that sensor drop-out will result in signal attenuation sufficient to cross the low glucose alarm threshold, which results in the presentation of an alarm condition to the user.

(93) Under closed-loop conditions, where the presence of a closed-loop insulin delivery algorithm offers the additional hazard mitigation of the automatic reduction or suspension of insulin delivery based upon real-time glucose level measurements (or based upon the recent history of glucose readings), an alternative low glucose alarm approach may be preferred. In this embodiment, the low glucose level alarm is triggered when a low glucose threshold is exceeded for a clinically significant duration.

(94) Various embodiments of such a delayed low glucose alarm are possible. For example, the alarm may be programmed to sound (1) after X minutes of continuous glucose measurements had been detected below some critical glucose value Y, where Xi=O; Yi=40 mg/dl; where X2=30; Y2=45 mg/dl; and where X3=120; Y3=60 mg/dl; or (2) After a continuous glucose measurement had been detected below some critical threshold A and had not recovered to either that critical threshold A or some intermediary threshold B within some specified duration (A′ and B′); or (3) After some cumulative measure of time and duration below a low-glucose threshold, such as when the integral of some low glucose threshold minus the glucose profile exceeds a specified value before the continuous glucose level measurements have recovered to some specified value; or (4) After X minutes of closed loop insulin delivery rate (or total volume administered) below some specified threshold. In this case, insulin delivered is treated as a surrogate for measured glucose level due to the anticipated function of the control algorithm programmed into the software that is operating in the controller and/or pump.

(95) Such an alarm approach is likely to reduce the frequency of false low glucose alarms by allowing for the possibility of the complete or partial recovery from the sensor-drop out to occur prior to the alarm presentation to the user. This loosening of alarm conditions is enabled by the increased safety to the user associated with the reduced or discontinued insulin infusion upon low

continuous glucose values that is offered by a close-loop insulin level algorithm.

(96) In another one embodiment of the invention, critical glucose level values are used to define the time allowed before the alarm sounds, resulting in a delayed alarm. For example, the time allowed value could be any appropriate function of glucose level, such as, for example only, and not limited to, a linear relationship.

(97) For example, in one embodiment, the processor re-evaluates the time allowed before the alarm is sounded at every newly received glucose level measurement which is below some predefined or user set value. Using such an approach, if the glucose level is at 60 mg/dL (and previously above 60 mg/dL), then the time until the alarm sounded would be determined to be 60 minutes. One minute later, when the next glucose level value is calculated, the "time until alarm sound" may be calculated to be the lesser of 59 minutes, that is, 60 minutes decremented by one, and the time associated with a linear function; which may be, for example, 3 minutes for every 1 mg/dL that the glucose value is below 60 mg/dL. Thus, if the second glucose value is 55 mg/dL, a decrease in glucose of 15 mg/dL, then the new "time until alarm sound" will be 45 minutes (60 minutes-(3 minutes per mg/dL.times.15 mg/dL). This determination is repeated for every glucose value received until the alarm sounds or the user's glucose level is greater than 60 mg/dL. Various approaches to this problem may be used individually or in combination in order to minimize false hypoglycemic alarms due to drop outs.

(98) Yet another embodiment of the present invention utilizes model based variable risk determinations to calculate an alarm delay to prevent the annunciation of false alarms, while still providing acceptable protection against a user entering an unacceptable glucose level condition. When CGM systems are used by persons with good glucose control, either by fully manual means, fully automated closed loop operation, or any hybrid means in between, the users glucose will typically be controlled much closer to the euglycemia range than would otherwise occur. This close control, coupled with the expected error distribution of CGM measurements, means that there will be an increased likelihood of CGM values close to or below the hypoglycemia limit. In addition to CGM calibration error, certain CGM signal distortion phenomena such as night time dropouts also increase the likelihood of an underreported glucose value by the CGM system.

(99) The combination of the three aforementioned factors interacts with standard hypoglycemia detectors to produce a higher incidence of false alarm rates. One embodiment of the invention designed to deal with this problem has been described above. Other approaches are also possible, and intended to be within the scope of the present invention, such as raising the importance and persistency of alarms depending on the length of time the alarm condition has existed without action by the user. Such approaches allow for false hypoglycemic alarms due to a combination of the three aforementioned factors to be minimized.

(100) There will, however, be instances where the patient's physiology and activity temporally pushes their glucose profile out of euglycemia and into either a higher likelihood of hypoglycemia or hyperglycemia. Examples of events where hyperglycemia is more likely would be after exercise, of after a meal where the user failed to receive an insufficient prandial bolus of insulin.

(101) If the user has a system including a CGM and an insulin delivery pump, information from the devices can be pooled or shared, and a model-based monitoring system can be used to modify the alarm mechanism to more efficiently minimize false alarms without imposing unnecessary risk to the patient. Using the information available from the CGM glucose monitor and insulin delivery system, as well as other information entered by the user whenever available (for example, the amount of exercise, state of health, and the like), a model-based system defined by appropriate software commands running on a processor of a controller can perform a prediction of glucose in terms of best estimate and upper/lower bounds for the present time up to a finite horizon in the future. Such a system can be achieved, for example, by implementing the user's model in the form of a Kalman Filter framework, where both the best estimate and variance of each model state is estimated, predicted, and modeled.

(102) Using the present-to-near-future best estimate and bounds, the user's most likely temporal range can be inferred. For example, assume the hypoglycemic detection mechanism of the programming of the controller or pump has 3 levels: 1) wait 30 minutes when the CGM determined glucose level crosses a 60 mg/dL threshold; 2) wait 15 minutes when the CGM determined glucose level crosses a 50 mg/dL threshold, and 3) alarm immediately when the CGM glucose level crosses a 45 mg/dL threshold. A Kalman framework may be used to determine the likelihood for a predicted future glucose level, given the user's current CGM glucose level and other insulin delivery history. This mechanism is implemented “as is” when the model's temporal glucose range suggests the lowest hypoglycemic likelihood. As the likelihood reaches the middle-range risk, then the mechanism is implemented with a 50% shortening of the pre-set delay for sounding the hypoglycemic alarm. For example, the detection levels set forth above may be modified as follows: 1′) wait 15 minutes when the CGM glucose level crosses the 60 mg/dL threshold; 2′) wait 7.5 minutes when the CGM glucose level crosses the 50 mg/dL threshold, and 3′) alarm immediately when the CGM glucose level crosses the 45 mg/dL threshold. When the highest risk of hypoglycemic likelihood is determined, the alarm threshold has 0 delay.
(103) In another embodiment, the same mechanism is implemented, that is: a) wait 30 minutes when the CGM glucose level crosses the 60 mg/dL threshold; b) wait 15 minutes when the CGM glucose level crosses the 50 gm/dL threshold; and 3) alarm immediately when the CGM glucose level crosses the 45 mg/dL threshold) when the model’ temporal glucose range suggests the lowest hypoglycemic likelihood. In the middle-range risk, the same delays are retained, but the threshold values are increased, such as, for example: a′) wait 30 minutes when the CGM glucose level crosses the 60+5 mg/dL threshold; b′) wait 15 minutes when the CGM glucose level crosses the 50+5 mg/dL threshold; and c′) alarm immediately when the CGM glucose level crosses the crosses 45+5 mg/dL threshold.
(104) In yet another embodiment, a hybrid of the two previous embodiments may be used. In such an embodiment, both the threshold values and the delay time may be adjusted based on the temporal glucose range determined by the model.
(105) The same types of mechanisms may be applied to hyperglycemia detection, with the tiered thresholds increasing in the order of threshold values. For example, where the CGM glucose level crosses a 180 mg/dL threshold the longest delay time is implemented; where the CGM glucose level crosses a 200 mg/dL threshold, a shorter delay time is implemented before sounding an alarm; and where the CGM glucose level crosses a 220 gm/dL threshold, an even shorter delay time is implemented, up to a maximum threshold with a zero delay time.
(106) While several specific embodiments of the invention have been illustrated and described, it will be apparent that various modifications can be made without departing from the spirit and scope of the invention. Accordingly, it is not intended that the invention be limited, except as by the appended claims.

Claims

1. A continuous glucose monitoring system comprising: a glucose monitoring medical device configured to be coupled externally to a body of a user, the glucose monitoring medical device comprising: a glucose sensor configured for implantation into the body of the user and for detecting a glucose level of the user, a processor, and a wireless communication unit communicatively coupled with and configured to wirelessly communicate with an electronic device of the system; and an electronic device comprising: a processor, a wireless communication unit communicatively coupled with and configured to wirelessly communicate with the wireless communication unit of the glucose monitoring medical device, and a memory operably coupled to the processor, wherein the memory includes instructions stored therein that, when executed by the processor, cause the processor to: receive a measurement of the glucose level of the user from the glucose monitoring

medical device; determine that the glucose level of the user has crossed a glucose threshold associated with a glucose alarm; delay an initial output of the glucose alarm by the electronic device for a delay time, wherein the delay time comprises a duration of time after the determination that the glucose level has crossed the glucose threshold; and cause the initial output of the glucose alarm after the delay time if the glucose level exceeds the glucose threshold throughout the delay time.

2. The system of claim 1, wherein the glucose threshold is settable by the user.

3. The system of claim 1, wherein the system is configured such that the glucose threshold is preprogrammed to the memory of the electronic device.

4. The system of claim 1, wherein the system is configured such that the glucose threshold is associated with hyperglycemia.

5. The system of claim 1, wherein the electronic device further comprises a display incorporating a graphical user interface, wherein the memory comprises instructions that, when executed by the processor of the electronic device, cause the processor to display the glucose alarm in the graphical user interface after the delay time.

6. The system of claim 5, wherein the display is a touch screen capable of responding to user activation.

7. The system of claim 1, wherein the continuous glucose monitoring system further comprises a remote device in wireless communication with the electronic device.

8. The system of claim 7, wherein the remote device is in further wireless communication with the glucose monitoring medical device through the electronic device.

9. The system of claim 1, wherein the duration of time is an annunciation delay time and the initial output of the of the glucose alarm is an annunciation of the glucose alarm.

10. The system of claim 1, wherein the duration of time of the delay time is a first duration of time, the glucose threshold is a first glucose threshold, and the memory includes instructions that, when executed by the processor, cause the processor of the electronic device to: determine that the glucose level of the user has crossed a second glucose threshold associated with the glucose alarm, wherein the second glucose threshold is greater than the first glucose threshold; and delay the output of the glucose alarm for a second delay time, wherein the second delay time comprises a second duration of time, wherein the second duration of time is shorter than the first duration of time.

11. The system of claim 1, wherein the duration of time of the delay time is a first duration of time, the glucose threshold is a first glucose threshold, and the memory includes instructions that, when executed by the processor, cause the processor of the electronic device to: determine that the glucose level of the user has crossed a second glucose threshold associated with the glucose alarm, wherein the second glucose threshold is greater than the first glucose threshold; and cause an output of the glucose alarm without delay upon determination that the glucose level has crossed the second glucose threshold.

12. The system of claim 1, wherein the memory includes instructions that, when executed by the processor of the electronic device, cause the processor to: provide a recommendation to control the glucose level of the user, wherein the recommendation is based on the detected glucose level.

13. The system of claim 12, wherein the recommendation includes the user ingesting an amount of carbohydrates.

14. The system of claim 12, wherein the recommendation includes reducing a basal rate of a medication delivery device.

15. The system of claim 12, wherein the recommendation includes increasing a time horizon of a medication delivery device of the user.

16. A method of delaying output of a glucose alarm in a continuous glucose monitoring system, the method comprising: detecting, by one or more processors of an electronic device and using a glucose sensor of a glucose monitoring medical device, a glucose level of a user, wherein the

electronic device is communicatively coupled with the glucose monitoring medical device, the glucose monitoring medical device is configured to be coupled externally to a body of the user, and the glucose sensor is configured for implantation into the body of the user; determining, by the one or more processors, that the glucose level of the user has crossed a glucose threshold associated with the glucose alarm; delaying, by the one or more processors, an initial output of the glucose alarm for a delay time, wherein the delay time comprises a duration of time after the determination that the glucose level has crossed the glucose threshold; and causing, by the one or more processors, the initial output of the glucose alarm after the delay time if the glucose level exceeds the glucose threshold throughout the delay time.
17. The method of claim 16, further comprising receiving, by the one or more processors, a value of the glucose threshold from the user.
18. The method of claim 16, wherein the electronic device is configured such that the glucose threshold is preprogrammed to the memory of the electronic device.
19. The method of claim 16, wherein the electronic device is configured such that the glucose threshold is associated with hyperglycemia.
20. The method of claim 16, wherein the electronic device further comprises a display incorporating a graphical user interface, and the method further comprises displaying the glucose alarm in the graphical user interface after the delay time.
21. The method of claim 16, wherein the duration of time is an annunciation delay time and the initial output of the of the glucose alarm is an annunciation of the glucose alarm.
22. The method of claim 16, wherein the duration of time of the delay time is a first duration of time, the glucose threshold is a first glucose threshold, and the method further comprises: determining, by the one or more processors, that the glucose level of the user has crossed a second glucose threshold associated with the glucose alarm, wherein the second glucose threshold is greater than the first glucose threshold; and delaying, by the one or more processors, the output of the glucose alarm for a second delay time, wherein the second delay time comprises a second duration of time, wherein the second duration of time is shorter than the first duration of time.
23. The method of claim 16, wherein the duration of time of the delay time is a first duration of time, the glucose threshold is a first glucose threshold, and the method further comprises: determining, by the one or more processors, that the glucose level of the user has crossed a second glucose threshold associated with the glucose alarm, wherein the second glucose threshold is greater than the first glucose threshold; and causing, by the one or more processors, an output of the glucose alarm without delay upon determination that the glucose level has crossed the second glucose threshold.
24. The method of claim 16, further comprising: providing, by the one or more processors, a recommendation to control the glucose level of the user, wherein the recommendation is based on the detected glucose level.
25. The method of claim 24, wherein the recommendation includes the user ingesting an amount of carbohydrates.
26. A non-transitory computer-readable medium comprising instructions which, when executed by one or more processors of an electronic device, are configured to cause the one or more processors to: detect, using a glucose sensor of a glucose monitoring medical device, a glucose level of a user, wherein the electronic device is communicatively coupled with the glucose monitoring medical device, the glucose monitoring medical device is configured to be coupled externally to a body of the user, and the glucose sensor is configured for implantation into the body of the user; determine that the glucose level of the user has crossed a glucose threshold associated with the glucose alarm; delay an initial output of the glucose alarm for a delay time, wherein the delay time comprises a duration of time after the determination that the glucose level has crossed the glucose threshold; and cause the initial output of the glucose alarm after the delay time if the glucose level exceeds the glucose threshold throughout the delay time.

27. The non-transitory computer-readable medium of claim 26, wherein the instructions are further configured to cause the one or more processors of the electronic device to receive a value of the glucose threshold from the user.
28. The non-transitory computer-readable medium of claim 26, wherein the electronic device is configured such that the glucose threshold is preprogrammed to the memory of the electronic device.
29. The non-transitory computer-readable medium of claim 26, wherein the electronic device is configured such that the glucose threshold is associated with hyperglycemia.
30. The non-transitory computer-readable medium of claim 26, wherein the electronic device further comprises a display incorporating a graphical user interface and wherein the instructions are further configured to cause the one or more processors to display the glucose alarm in the graphical user interface after the delay time.